

HAT-P-27b

R-0106

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_250208 · created 2026-07-16T03:37:07+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460711.9684 +/- 0.0078 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.137 +/- 0.037
Transit depth (Rp/Rs)^2	1.87 +/- 1.01 [%]
Orbital Inclination (inc)	85.29 +/- 2.36
Airmass coefficient 1 (a1)	1.053 +/- 0.011
Airmass coefficient 2 (a2)	-0.039 +/- 0.04
Scatter in the residuals of the lightcurve fit is	1.03 %
Best Comparison Star	#3 - [564, 212]
Optimal Aperture	3.28
Optimal Annulus	6.42
Transit Duration (day)	0.027 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.137 $\pm$ 0.037 vs 0.119 (deviation 0.49 σ)
Duration fit vs published	0.027 d vs 0.07 d (deviation 2.69 σ)
Mid-time O-C	-4383.2 min
Depth SNR	3.7
Residual scatter	1.03 %

Light curve

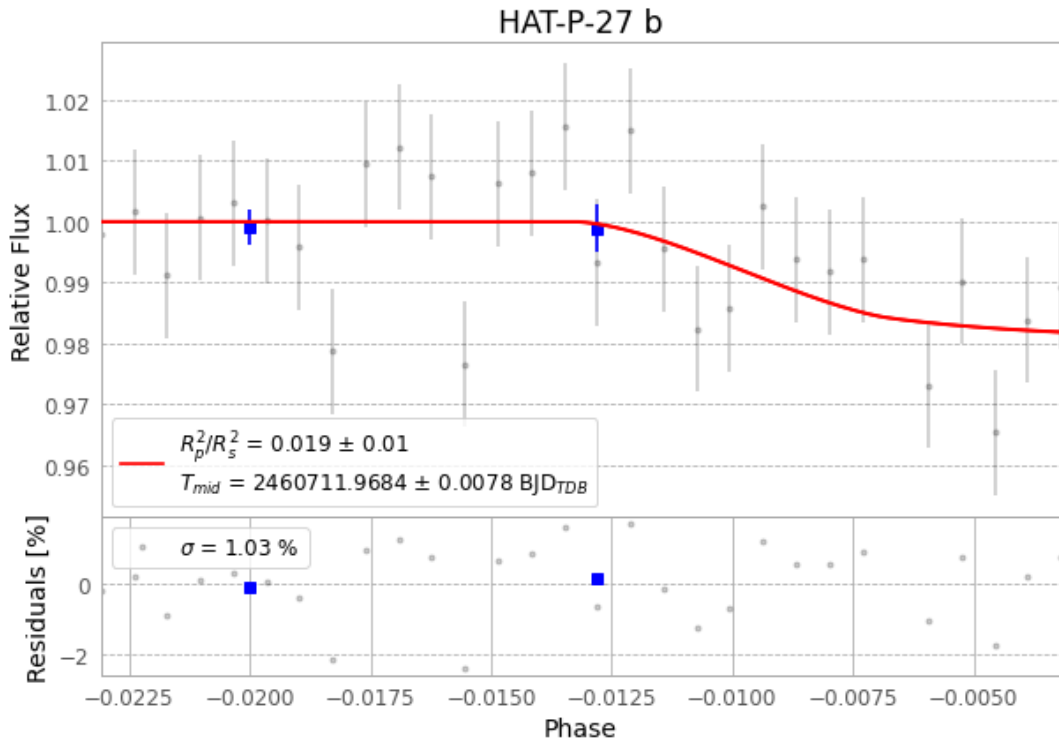


Figure 1 — FinalLightCurve_HAT-P-27 b_2025-02-08.png (SHA-256 242896e1d6b4185b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [344, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 14250d57636fd8bf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

30 FITS frames (20 MB), HATP-27250208102715.FITS → HATP-27250208115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-250208.log (de149d2e5846...), FinalLightCurve_HAT-P-27 b_2025-02-08.png (242896e1d6b4...), FinalLightCurve_HAT-P-27 b_2025-02-08.csv (27b592c19f7b...)

Compute resources

Wall time 100.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 03:37Z.